

Brains

What are they?

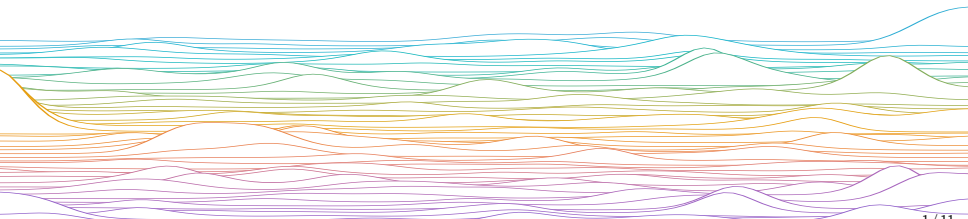
Computational Neuroscience

University of Bristol

M Rule

Learning outcomes:

- ▶ Recall key facts, figures, statistics, and anatomical terms from these slides
- ▶ Describe some of the specialized, interacting sub-systems that constitute the brain & their function.

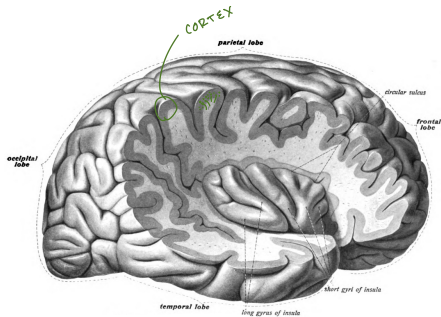


"Grey" matter

- ▶ Neuron cell bodies, local connections
- ▶ Cortex, nuclei

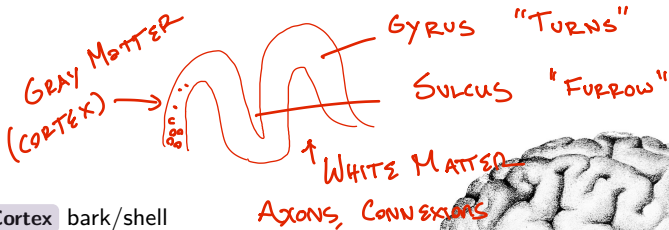
"White" matter

- ▶ Long-range electrical wiring (axons)
 - ▶ Myelin insulation (oligodendroglia)
- LIPID BASED*



Dr. Johannes Sobotta (c. 1908)

Textbook and Atlas of Human Anatomy, (via Wikimedia)



Cortex bark/shell

3–6 layers (6 layers ~ neocortex)

Larger brains

- ▶ More functionally specialized
- ▶ Folded:
 - Gyri (hills)
 - Sulci (valleys)
- ▶ Humans: $\approx 0.25 \text{ m}^2$ total area

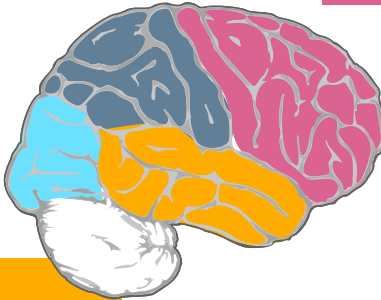


Outer surface of the human brain,
Sanger Brown, c. 1894 (via Wikimedia)

Parietal
Multisensory Integration
Abstract workspace
Somatosensation

Frontal
Movement (language: Broca's area)
Executive Function, Planning
Emotional Regulation

Occipital
Vision



Temporal
Auditory,
Semantic Memory
Language: Wernicke's area

FUNCTIONAL SPECIALIZATION
PARTLY INFLUENCED BY
ANATOMICAL CONNECTIONS,
SOME DEVELOPMENTAL
PLASTICITY, BUT THIS HAS LIMITS.

Parietal
Multisensory Integration
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PRIMARY MOTOR
← MORE MOTOR-
LIKE

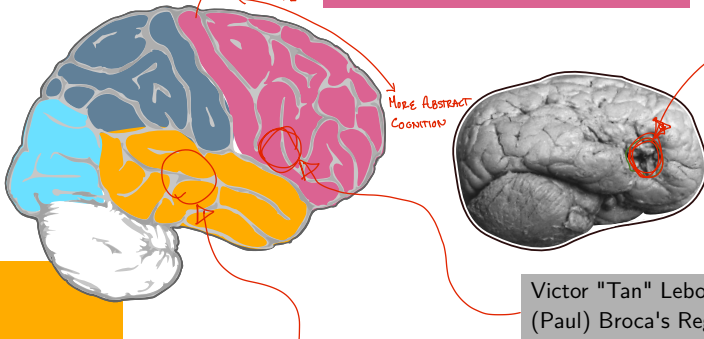
MORE ABSTRACT
COGNITION

BROCA'S
AREA

WERNICKE'S AREA

Victor "Tan" Leborgne's brain
(Paul) Broca's Region
Expressive aphasia

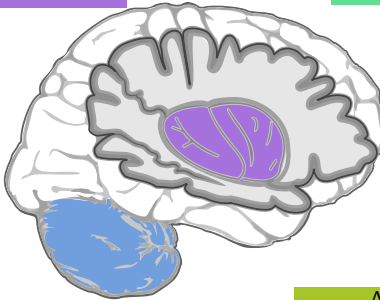
Circle Broca's area, mention "Tan"



Insula
Taste, smell, craving
Interoception, emotion

Hippocampus
Memory, navigation

Week 5

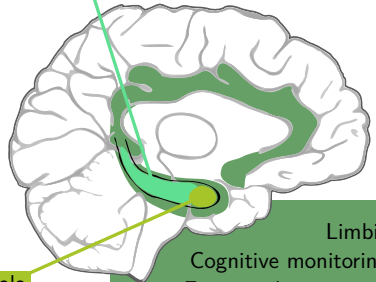


Cerebellum
Skilled movement
Timing, prediction

Week 4

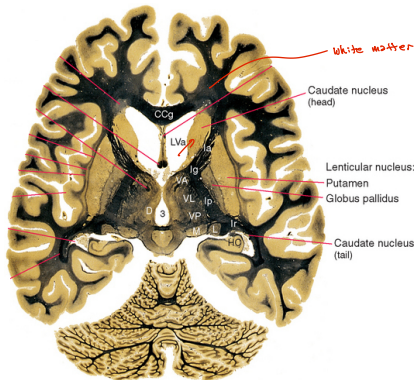
Amygdala
Fear conditioning

Limbic
Cognitive monitoring
Emotional cognition
Memory encoding/retrieval



Basal ganglia

- ▶ Deep-brain nuclei
- ▶ Choose/decide/start/stop
- ▶ Reward-based learning

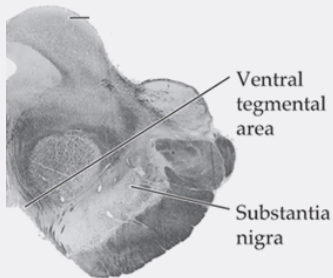


Nolte & Angevine (2007)

Parkinson's

Substantia nigra degeneration

- ▶ Tremor/rigidity/slowness
- ▶ Cognition/sleep



J. Martin Neuroanatomy Text and Atlas

end

